Remove Relationships

Now let's say that **Spider-Boy** tells **Rusty-Man** something like:

> I don't feel so good Mr. Sharp

And then for some reason needs to leave the **Preventers** for some years. 😭

We can remove the relationship by setting it to `None`, the same as with the `team_id`, it also works with the new relationship attribute `.team`:

Python 3.10+

```python
# Code above omitted 👆

def update_heroes():
    with Session(engine) as session:
        statement = select(Hero).where(Hero.name == "Spider-Boy")
        result = session.exec(statement)
        hero_spider_boy = result.one()

        hero_spider_boy.team = None
        session.add(hero_spider_boy)
        session.commit()

        session.refresh(hero_spider_boy)
        print("Spider-Boy without team:", hero_spider_boy)

# Code below omitted 👇
```

👀 **Full file preview**

Python 3.10+

```python
from sqlmodel import Field, Relationship, Session, SQLModel, create_engine, select


class Team(SQLModel, table=True):
    id: int | None = Field(default=None, primary_key=True)
    name: str = Field(index=True)
    headquarters: str

    heroes: list["Hero"] = Relationship(back_populates="team")


class Hero(SQLModel, table=True):
    id: int | None = Field(default=None, primary_key=True)
    name: str = Field(index=True)
    secret_name: str
    age: int | None = Field(default=None, index=True)

    team_id: int | None = Field(default=None, foreign_key="team.id")
    team: Team | None = Relationship(back_populates="heroes")


sqlite_file_name = "database.db"
sqlite_url = f"sqlite:///{sqlite_file_name}"

engine = create_engine(sqlite_url, echo=True)


def create_db_and_tables():
    SQLModel.metadata.create_all(engine)


def create_heroes():
    with Session(engine) as session:
        team_preventers = Team(name="Preventers", headquarters="Sharp Tower")
        team_z_force = Team(name="Z-Force", headquarters="Sister Margaret's Bar")

        hero_deadpond = Hero(
            name="Deadpond", secret_name="Dive Wilson", team=team_z_force
        )
        hero_rusty_man = Hero(
            name="Rusty-Man", secret_name="Tommy Sharp", age=48, team=team_preventers
        )
        hero_spider_boy = Hero(name="Spider-Boy", secret_name="Pedro Parqueador")
        session.add(hero_deadpond)
        session.add(hero_rusty_man)
        session.add(hero_spider_boy)
        session.commit()

        session.refresh(hero_deadpond)
        session.refresh(hero_rusty_man)
        session.refresh(hero_spider_boy)

        print("Created hero:", hero_deadpond)
        print("Created hero:", hero_rusty_man)
        print("Created hero:", hero_spider_boy)

        hero_spider_boy.team = team_preventers
        session.add(hero_spider_boy)
        session.commit()
        session.refresh(hero_spider_boy)
        print("Updated hero:", hero_spider_boy)

        hero_black_lion = Hero(name="Black Lion", secret_name="Trevor Challa", age=35)
        hero_sure_e = Hero(name="Princess Sure-E", secret_name="Sure-E")
        team_wakaland = Team(
            name="Wakaland",
            headquarters="Wakaland Capital City",
            heroes=[hero_black_lion, hero_sure_e],
        )
        session.add(team_wakaland)
```

```python
        session.commit()
        session.refresh(team_wakaland)
        print("Team Wakaland:", team_wakaland)

        hero_tarantula = Hero(name="Tarantula", secret_name="Natalia Roman-on", age=32)
        hero_dr_weird = Hero(name="Dr. Weird", secret_name="Steve Weird", age=36)
        hero_cap = Hero(
            name="Captain North America", secret_name="Esteban Rogelios", age=93
        )

        team_preventers.heroes.append(hero_tarantula)
        team_preventers.heroes.append(hero_dr_weird)
        team_preventers.heroes.append(hero_cap)
        session.add(team_preventers)
        session.commit()
        session.refresh(hero_tarantula)
        session.refresh(hero_dr_weird)
        session.refresh(hero_cap)
        print("Preventers new hero:", hero_tarantula)
        print("Preventers new hero:", hero_dr_weird)
        print("Preventers new hero:", hero_cap)


def select_heroes():
    with Session(engine) as session:
        statement = select(Team).where(Team.name == "Preventers")
        result = session.exec(statement)
        team_preventers = result.one()

        print("Preventers heroes:", team_preventers.heroes)


def update_heroes():
    with Session(engine) as session:
        statement = select(Hero).where(Hero.name == "Spider-Boy")
        result = session.exec(statement)
        hero_spider_boy = result.one()

        hero_spider_boy.team = None
        session.add(hero_spider_boy)
        session.commit()

        session.refresh(hero_spider_boy)
        print("Spider-Boy without team:", hero_spider_boy)


def main():
    create_db_and_tables()
    create_heroes()
    select_heroes()
    update_heroes()


if __name__ == "__main__":
    main()
```

And of course, we should remember to add this `update_heroes()` function to `main()` so that it runs when we call this program from the command line:

Python 3.10+

```python
# Code above omitted 👆

def main():
    create_db_and_tables()
    create_heroes()
    select_heroes()
    update_heroes()

# Code below omitted 👇
```

👀 Full file preview

Python 3.10+

```python
from sqlmodel import Field, Relationship, Session, SQLModel, create_engine, select


class Team(SQLModel, table=True):
    id: int | None = Field(default=None, primary_key=True)
    name: str = Field(index=True)
    headquarters: str

    heroes: list["Hero"] = Relationship(back_populates="team")


class Hero(SQLModel, table=True):
    id: int | None = Field(default=None, primary_key=True)
    name: str = Field(index=True)
    secret_name: str
    age: int | None = Field(default=None, index=True)

    team_id: int | None = Field(default=None, foreign_key="team.id")
    team: Team | None = Relationship(back_populates="heroes")


sqlite_file_name = "database.db"
sqlite_url = f"sqlite:///{sqlite_file_name}"

engine = create_engine(sqlite_url, echo=True)


def create_db_and_tables():
    SQLModel.metadata.create_all(engine)


def create_heroes():
    with Session(engine) as session:
        team_preventers = Team(name="Preventers", headquarters="Sharp Tower")
        team_z_force = Team(name="Z-Force", headquarters="Sister Margaret's Bar")

        hero_deadpond = Hero(
            name="Deadpond", secret_name="Dive Wilson", team=team_z_force
        )
        hero_rusty_man = Hero(
            name="Rusty-Man", secret_name="Tommy Sharp", age=48, team=team_preventers
        )
        hero_spider_boy = Hero(name="Spider-Boy", secret_name="Pedro Parqueador")
        session.add(hero_deadpond)
        session.add(hero_rusty_man)
        session.add(hero_spider_boy)
        session.commit()

        session.refresh(hero_deadpond)
        session.refresh(hero_rusty_man)
        session.refresh(hero_spider_boy)

        print("Created hero:", hero_deadpond)
        print("Created hero:", hero_rusty_man)
        print("Created hero:", hero_spider_boy)

        hero_spider_boy.team = team_preventers
        session.add(hero_spider_boy)
        session.commit()
        session.refresh(hero_spider_boy)
        print("Updated hero:", hero_spider_boy)

        hero_black_lion = Hero(name="Black Lion", secret_name="Trevor Challa", age=35)
        hero_sure_e = Hero(name="Princess Sure-E", secret_name="Sure-E")
        team_wakaland = Team(
            name="Wakaland",
            headquarters="Wakaland Capital City",
            heroes=[hero_black_lion, hero_sure_e],
        )
        session.add(team_wakaland)
```

```python
        session.commit()
        session.refresh(team_wakaland)
        print("Team Wakaland:", team_wakaland)

        hero_tarantula = Hero(name="Tarantula", secret_name="Natalia Roman-on", age=32)
        hero_dr_weird = Hero(name="Dr. Weird", secret_name="Steve Weird", age=36)
        hero_cap = Hero(
            name="Captain North America", secret_name="Esteban Rogelios", age=93
        )

        team_preventers.heroes.append(hero_tarantula)
        team_preventers.heroes.append(hero_dr_weird)
        team_preventers.heroes.append(hero_cap)
        session.add(team_preventers)
        session.commit()
        session.refresh(hero_tarantula)
        session.refresh(hero_dr_weird)
        session.refresh(hero_cap)
        print("Preventers new hero:", hero_tarantula)
        print("Preventers new hero:", hero_dr_weird)
        print("Preventers new hero:", hero_cap)


def select_heroes():
    with Session(engine) as session:
        statement = select(Team).where(Team.name == "Preventers")
        result = session.exec(statement)
        team_preventers = result.one()

        print("Preventers heroes:", team_preventers.heroes)


def update_heroes():
    with Session(engine) as session:
        statement = select(Hero).where(Hero.name == "Spider-Boy")
        result = session.exec(statement)
        hero_spider_boy = result.one()

        hero_spider_boy.team = None
        session.add(hero_spider_boy)
        session.commit()

        session.refresh(hero_spider_boy)
        print("Spider-Boy without team:", hero_spider_boy)


def main():
    create_db_and_tables()
    create_heroes()
    select_heroes()
    update_heroes()


if __name__ == "__main__":
    main()
```

Recap

This chapter was too short for a recap, wasn't it? 🤔

Anyway, **relationship attributes** make it easy and intuitive to work with relationships stored in the database. 🎉